CSE2104 Offline-3

1. Given an array of **N** integers, simulate Quick Sort using the **last element as pivot** and count the number of comparisons performed during the sorting process.
2. Given an array and a pivot, rearrange the array such that elements less than pivot come before it and greater come after.
3. Count the number of swaps required to sort an array using quick sort.
4. Output the pivot chosen at each recursive call in quick sort (always pick the last element as pivot).
5. For each recursive quick sort call, output the final index where pivot is placed.
6. For each pivot selected in quick sort (last element as pivot), count how many elements are less than the pivot at that step.
7. Given two sorted arrays, find the median of the combined sorted array using merge sort. Also print the merged array.
8. Calculate the total number of comparisons made during merge sort.
9. Count the total number of merge operations performed when sorting an array using merge sort.
10. Determine the total number of recursive calls merge sort makes for a given array.

Prepare proper sample input output for each problem. Also include your prepared sample input output to the pdf report along with the problem statement.

Submit 10 **.cpp** files and a pdf report (Sample Report). Follow the naming convention for the .cpp file: **StudentID_Offline3_Task#.cpp** and the report: **StudentID_Offline3.pdf**. For example: 180104072_Offline3_Task1.cpp and 180104072_Offline3.pdf.

Follow the submission instructions and deadline strictly.